

Algorithm and Data Structure

TD5 : Problem Solving with Dynamic Programming

Objectives

Developing Dynamic Programming algorithms to solve a given problem

1 Strasbourg's Christmas Light Plan

The city of Strasbourg, famous for its magical Marché de Noël, is planning a breathtaking street display of Christmas lights connecting the lamp posts on both sides of the main avenue. There are N lamp posts on the left side of the street and N lamp posts on the right side.

Each proposed light string connects a lamp post on the left to one on the right, represented as a pair $(left, right)$ where these two values represent the position of the lamp posts.

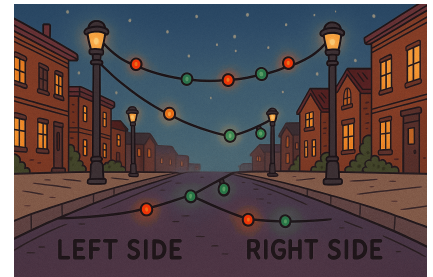
However, some strings cross each other, creating a tangled mess and the city wants the display to be elegant, not chaotic!

“Note“ : Two light strings $(left_1, right_1)$ and $(left_2, right_2)$ are said to cross if the following condition holds :

$$left_1 < left_2 \text{ and } right_1 > right_2$$

or vice versa :

$$left_1 > left_2 \text{ and } right_1 < right_2$$



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Task

Your job is to help Strasbourg choose :

1. The maximum number of light strings that can be hung without crossing, and
2. Exactly which lamp pairs (left and right) should be used in this optimal design.

You may not change the endpoints of any proposed pair; you may only select or discard pairs. Each lamp post may be used by at most one selected string.

Input

- . First line : integer N representing the number of lamp posts in total
- . Followed by N lines with a pair of integers each $(left, right)$ where $left$ is the position of a lamp post on the left side of the street and $right$ represent the position of a lamp post on the right side of the street

Output

- . On the first line, print the maximum number of non-crossing lights
- . On the next lines, print the pairs $(left, right)$ used in the optimal configuration, in increasing order of the left lamp post

Constraints

- . $1 \leq N \leq 2 \cdot 10^5$
- . $1 \leq left, right \leq 10^9$

Example

Input :

```
5
6 2
4 3
2 6
1 5
3 4
```

Output :

```
2
1 5
2 6
```

2 Catch'em All!

You've entered the Safari Zone, a vast wild area filled with Pokémon! The zone is represented as a grid of size $n \times m$. Each area (i, j) contains a certain number of wild Pokémon you can try to catch.

You start from the entrance at the top-left corner and must reach the exit at the bottom-right corner.

From any cell (i, j) , you may only move one step down or one step right.

However, you only have k Poké Balls! You cannot catch more Pokémon than that, and you don't want to waste any Poké Balls either. Therefore, your goal is to plan a route such that you catch exactly k Pokémon by the time you reach the exit.

Task

Find the number of distinct paths from the entrance to the exit that allow you to catch exactly k Pokémon.

Input Format

- . The first line contains three integers n , m , and k ; the number of rows, the number of columns, and the total number of Poké Balls.
- . The next n lines each contain m integers each, where the element in the i_{th} line and j_{th} column represents the number of Pokémon available to be caught in area (i, j) .

Output Format

Output a single integer; the number of possible paths that result in catching exactly k Pokémon.

Constraints

- . $0 \leq k \leq 100$
- . $1 \leq n, m \leq 100$
- . $0 \leq grid[i][j] \leq 200$



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Example

Input

```
3 3 12
1 2 3
4 6 5
3 2 1
```

Output

```
2
```

Explanation

There are 2 possible paths with exactly 12 Pokémon, $(1 + 2 + 6 + 2 + 1)$ and $(1 + 2 + 3 + 5 + 1)$.

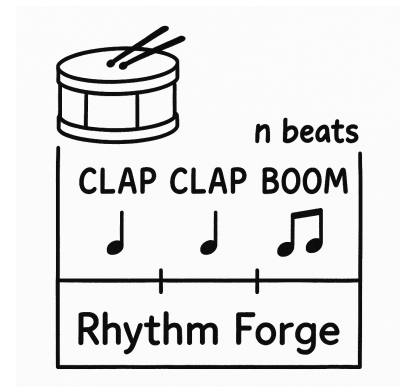
3 Rhythm Forge

You are designing beat patterns for a rhythm game. A pattern is a sequence of notes that fills exactly n beats.

- . A CLAP occupies exactly 1 beat.
- . A BOOM occupies exactly 2 beats.

You can use any combination of these notes, in any order, as long as the total duration adds up to exactly n beats.

Your task is to find how many distinct patterns can be formed. Since the number can be large, output it modulo 1 000 000 007.



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Task

Compute the number of distinct beat patterns of total length n . Let this number be C_n . You must print $C_n \bmod (10^9 + 7)$.

Helper : How to Expand Sums Under Modulo

When dealing with addition or multiplication under a modulus M :

$$(a + b) \bmod M = ((a \bmod M) + (b \bmod M)) \bmod M$$

$$(a \times b) \bmod M = ((a \bmod M) \times (b \bmod M)) \bmod M$$

Always take modulo after every addition or multiplication to prevent overflow.

Input Format

- . The first line contains an integer q , the number of queries.
- . Each of the next q lines contains a single integer n , the number of beats.

Output Format

For each query, output a single integer, the number of valid patterns modulo $(10^9 + 7)$.

Constraints

- . $1 \leq q \leq 2 \cdot 10^5$
- . $0 \leq n \leq 10^{18}$
- . Modulus $M = 10^9 + 7$

Example Input

```
6
0
1
2
3
4
10
```

Example Output

```
1
1
2
3
5
89
```

Explanation :

- . For $n = 3$: CLAP+CLAP+CLAP, CLAP+BOOM, BOOM+CLAP $\Rightarrow$ 3 ways.
- . For $n = 4$: 5 ways.
- . For $n = 10$: 89 ways.